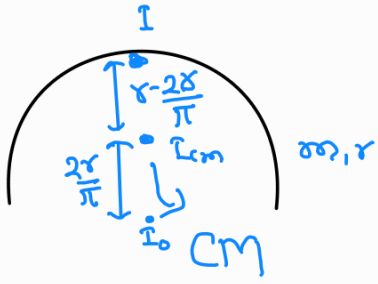
(24) Time period of Compound pendulum

$$T = 2\pi\sqrt{\frac{I}{mgu}}$$



Here,

$$I_o = mr^2$$

$$I_{cm} + m\left(\frac{2r}{\pi}\right)^2 = mr^2$$

$$I_{cm} = mr^2 - m\left(\frac{2r}{\pi}\right)^2$$

also, $I = I_{cm} + m\left(r - \frac{2r}{\pi}\right)^2$

$$= mr^2 - m\left(\frac{2r}{\pi}\right)^2 + mr^2 + m\left(\frac{2r}{\pi}\right)^2 - \frac{4mr^2}{\pi}$$

$$= 2mr^2 - \frac{4mr^2}{\pi}$$

$$= 2mr^2\left(1 - \frac{2}{\pi}\right)$$

$$T = 2\pi\sqrt{\frac{2mr^2\left(1-\frac{2}{\pi}\right)}{mgr\left(1-\frac{2}{\pi}\right)}} = 2\pi\sqrt{\frac{2r}{g}}$$

$$= 2\pi\sqrt{\frac{2\times \frac{20}{100}}{10}}$$

$$= 2\pi\sqrt{\frac{4}{100}}$$

$$= \frac{2\pi}{5}\ \text{Sec}$$

(25) $y = 2\text{cm}\ \sin(3t - 4u)$

$$v_y = \frac{dy}{dt} = 6\ \text{cm/s}\ \cos(3t - 4u)$$

(a) at $t=0$, $u=0$

$v_y = 6\cos(0-0) = +6$ cm/s

It moves toward the y-axis

(b) $\frac{2\pi}{\lambda} = 4 \Rightarrow \lambda = \frac{\pi}{2}$

$\Delta u = (8-1)$ m $= 7$ m

$\Delta\phi = \frac{2\pi}{\lambda}\Delta u = \frac{2\pi}{\pi/2} \times 7 = 28 \neq 2n\pi$

Not in phase

(c) at $u = \frac{\pi}{8}$ & $t = \frac{\pi}{6}$

$y = 2\sin\left(\frac{\pi}{6}\times 3 - \frac{\pi}{8}\times 4\right)$

$= 2\sin(0)$

$= 0$

Particle is at mean position, its acceleration is zero & now it will increase in magnitude

(d) velocity of particle is $\perp$ to direction of wave.

(26) $\Delta P = \frac{8\eta l}{\pi r^4} Q$

$\frac{\Delta(\Delta P)}{\Delta P} = \frac{\Delta l}{l} + 4\frac{\Delta r}{r}$

$= \frac{0.1}{10} + 4\times\frac{0.01}{2.00}$

$= 0.01 + 0.02$

$= 0.03$

$\frac{\Delta(\Delta P)}{\Delta P} \times 100\% = 0.03 \times 100\% = 3\%$

(27) $KE_{max} = \frac{hc}{\lambda} - \phi_0$

$\Rightarrow \frac{h^2}{2m_e\lambda_d^2} = \frac{hc}{\lambda} - \phi_0$

$\Rightarrow \frac{h^2}{2m_e}\left(-2\lambda_d^{-3}\Delta\lambda_d\right) = hc\left(-\lambda^{-2}\Delta\lambda\right) - 0$

$\Rightarrow \frac{\Delta\lambda_d}{\lambda_d^3} \propto \frac{\Delta\lambda}{\lambda^2}$

$\frac{\Delta\lambda_d}{\Delta\lambda} \propto \frac{\lambda_d^3}{\lambda^2}$

, $\lambda_d = \frac{h}{\sqrt{2m_e KE_{max}}}$

$\Rightarrow 2m_e K_{max} = \frac{h^2}{\lambda_d^2}$

$\Rightarrow K_{max} = \frac{h^2}{2m_e\lambda_d^2}$

(28) Let $E = E_0 \sin(\omega t - Ku)$

Conduction current density

$J_c = \sigma E = \sigma E_0 \sin(\omega t - Ku) \Rightarrow \langle J_c \rangle = \sigma E_0 \times \frac{2}{\pi}$

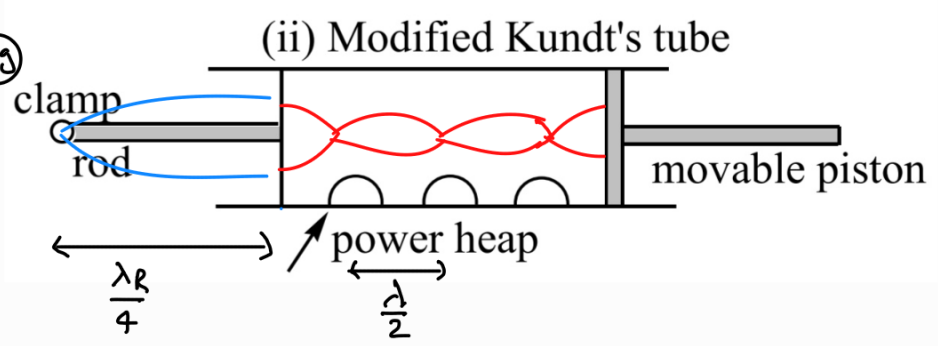
Displacement current

$i_d = \varepsilon\varepsilon_0 A \frac{dE}{dt}$

$J_d = \varepsilon\varepsilon_0 \frac{dE}{dt} = \varepsilon\varepsilon_0\omega E_0 \cos(\omega t - Ku) \Rightarrow \langle J_d \rangle = \varepsilon\varepsilon_0\omega E_0 \times \frac{2}{\pi}$

$\frac{\langle J_c \rangle}{\langle J_d \rangle} = \frac{\sigma E_0 \times \frac{2}{\pi}}{\varepsilon\varepsilon_0\omega E_0 \times \frac{2}{\pi}} = \frac{\sigma}{\varepsilon\varepsilon_0\omega}$

(29)



$\frac{\lambda}{2} = 0.6\,m \Rightarrow \lambda = 1.2\,m$

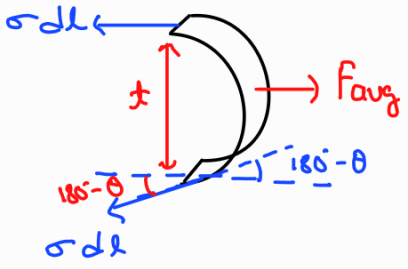
$v_s = 330\,m/s$

$f = \frac{v_s}{\lambda} = \frac{330}{1.2}\,Hz = 275\,Hz$

In rod, $\frac{\lambda_R}{4} = 1m \Rightarrow \lambda_R = 4m$

$v_R = \lambda_R f = 275 \times 4 \text{ m/s} = 1100 \text{ m/s}$

(30) take a small strip of length dl on side



$F_{avg} = \sigma dl + \sigma dl \cos(180° - \theta)$

$\left(\rho g \frac{t}{2}\right)(t\, dl) = \sigma\, dl - \sigma\, dl \cos\theta$

$\frac{\rho g t^2}{2} \cancel{dl} = \sigma \cancel{dl}(1 - \cos\theta)$

$t^2 = \frac{2\sigma(1-\cos\theta)}{\rho g}$

$t = \sqrt{\frac{2\sigma(1-\cos\theta)}{\rho g}}$

(31) (a) $V = h_1 \rho_1 = h_2 \rho_2$

$h_1 = \frac{V}{\rho_1}$, $h_2 = \frac{V}{\rho 2}$

$U_1 = h_1 - h_2 = \frac{V}{\rho_1} - \frac{V}{\rho_2} = V\left(\frac{1}{\rho_1} - \frac{1}{\rho_2}\right)$

(b) Pressure a same horizontal level is same in a static fluid. So both are at same level.

$y_1 = 0$

$$b\sqrt{\frac{2MgL(\sin\alpha-\sin\beta)}{mb^2+ML^2\sin^4\beta}}$$

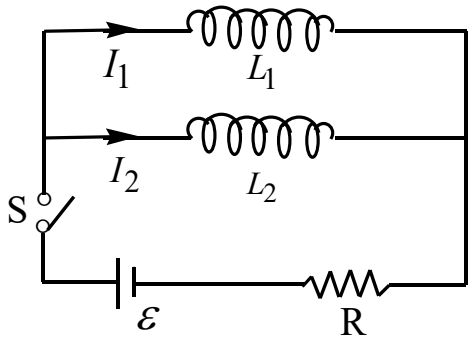
C) Angular speed of the rod when angle between rod and the floor is β is

$$\sqrt{\frac{2MgL(\sin\alpha-\sin\beta)}{mb^2\cos ec^4\beta+ML^2}}$$

D) Angular speed of the rod when angle between rod and the floor is β is

$$\sqrt{\frac{MgL(\sin\alpha-\sin\beta)}{mb^2+ML^2\cos^4\beta}}$$

33. In the circuit shown, the switch is closed at $t=0$ (L_1 and L_2 are ideal inductors and initial current at both inductors is zero)



A) Just after $t=0, I_1=I_2=0$

B) at any time $t=t, \frac{I_1}{I_2}=\frac{L_2}{L_1}$ ⇒ they are in parallel

$V_1=V_2 \Rightarrow \frac{d\Phi_1}{dt}=\frac{d\Phi_2}{dt} \Rightarrow \Phi_1=\Phi_2$

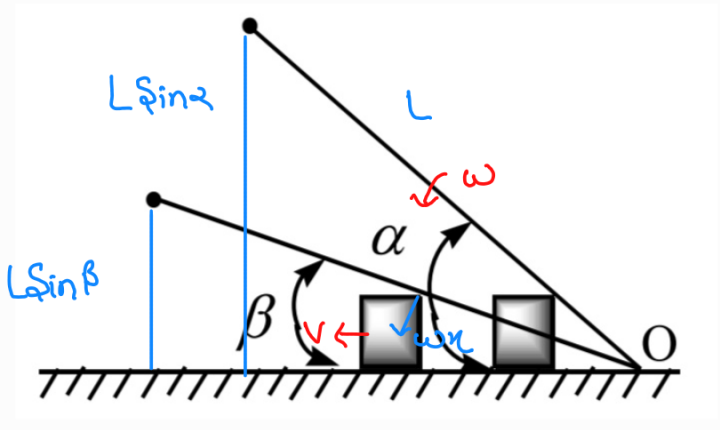
$\Rightarrow L_1I_1=L_2I_2$

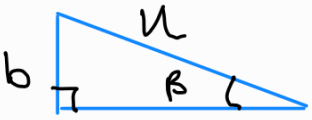
C) at any time $t=t, I_1+I_2=\frac{\varepsilon}{R}$

D) at $t=\infty$, current flowing through the resistor is $\frac{\varepsilon}{R}$. → behave as wire $R_1=R_2=0$

34. A solid sphere is held at rest on the fixed hemispherical bowl made of insulating material such that centre of sphere lies on the horizontal diameter as shown in figure. Total charge $q=5.1\times10^{-1}C$ and mass $m=42$ grams, both are distributed uniformly in the sphere. There exists unform magnetic field B in space as shown in the figure. consider contact surface to be sufficiently rough. Spherical bowl is made of non-magnetic material. Radius of bowl = 10 cm, radius of sphere 3 cm. $g=10\,m/s^2$. Now if sphere is released

(32)





$$\sin\beta = \frac{b}{x} \Rightarrow x = \frac{b}{\sin\beta}$$

$$\omega x = V\sin\beta$$

$$\omega \times \frac{b}{\sin\beta} = V\sin\beta$$

$$\omega = \frac{V}{b}\sin^2\beta$$

From COME

$$\Delta K = -\Delta U$$

$$\left(\frac{1}{2}I\omega^2 + \frac{1}{2}mV^2\right) - 0 = -\left(MgL\sin\beta - MgL\sin\alpha\right)$$

$$\frac{1}{2} \times ML^2 \times \omega^2 + \frac{1}{2}mV^2 = MgL\left(\sin\alpha - \sin\beta\right)$$

$$ML^2\omega^2 + mV^2 = 2MgL\left(\sin\alpha - \sin\beta\right)$$

$$ML^2 \times \frac{V^2}{b^2}\sin^4\beta + mV^2 = 2MgL\left(\sin\alpha - \sin\beta\right)$$

$$V^2\left(\frac{ML^2}{b^2}\sin^4\beta + m\right) = 2MgL\left(\sin\alpha - \sin\beta\right)$$

$$V = \sqrt{\frac{2MgL\,b^2\left(\sin\alpha - \sin\beta\right)}{ML^2\sin^4\beta + mb^2}}$$

$$\omega = \sqrt{\frac{2MgL\,b^2\left(\sin\alpha - \sin\beta\right)}{ML^2\sin^4\beta + mb^2}} \times \frac{\sin^2\beta}{b}$$

(34) Work done by B=0

From COME

$$K_i + P_i = K_f + P_f$$

$$0 + 0 = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 - mg(R-r)\sin\theta$$

$$\frac{1}{2}mv^2 + \frac{1}{2} \times \frac{2}{5}mr^2 \times \frac{v^2}{r^2} = mg(R-r)\sin\theta$$

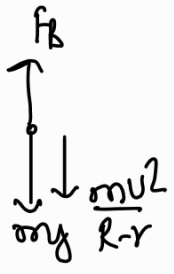
$$\frac{v^2}{2} \times \frac{7}{5} = g(R-r)\sin\theta$$

$$v = \sqrt{\frac{10}{7}g(R-r)\sin\theta}$$

For loosing Contact

$N = 0$

& For critical case $\theta = 90^\circ$, at the bottom



$$F_B = mg + \frac{mv^2}{R-r}$$

$$\left(v = \sqrt{\frac{10}{7}g(R-r)} \right)$$

$$= mg + \frac{m}{R-r} \times \frac{10}{7}g(R-r)$$

$$= mg + \frac{10mg}{7}$$

$$qvB = \frac{17}{7}mg$$

$$B = \frac{17}{7}\frac{mg}{qv}$$

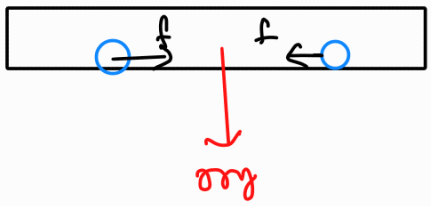
$$= \frac{17}{7} \times \frac{(42\times10^{-3}) \times 10}{(5.1\times10^{-1}) \times \sqrt{\frac{10}{7}\times10\times(10-3)\times10^{-2}}}$$

$$= \frac{17}{7} \times \frac{42\times10^{-1}}{5.1\times1}$$

$$= 2T$$

(35) (a) insects are doing work which will change energy of system

(b)



Net torque about centre is zero.

$\Rightarrow$ angular momentum about centre is conserved

(c) mg is external force $\Rightarrow$ Linear momentum not conserved

(d) $I\omega = \text{Const}$ but $I \neq \text{Const}$.

$\omega \neq$ const.

(36) time of flight, $t = \frac{2u\sin\theta}{g} = \frac{2 \times 10 \times \sin 30}{10} = 1$ sec

at any time t, from COAM about COM.

$$I_0\omega_0 = I'\omega$$

$$\frac{ML^2}{12}\omega_0 = \left(\frac{ML^2}{12} + 2m\left(\frac{1}{2}\times t\right)^2\right)\omega$$

$$\frac{1\times 1^2}{12} \times 8 = \left(\frac{1\times 1^2}{12} + 2\times\frac{1}{6}\times\frac{1}{4}\times t^2\right)\omega$$

$$8 = (1+t^2)\omega$$

$$\omega = \frac{8}{1+t^2}$$

$$\frac{d\theta}{dt} = \frac{8}{1+t^2}$$

$$\int_0^{\theta_0} d\theta = 8\int_0^T \frac{1}{1+t^2}\,dt$$

$$\theta_0 = 8\left[\tan^{-1}(t)\right]_0^T$$

$$= 8\left[\tan^{-1}(1) - \tan^{-1}(0)\right]$$

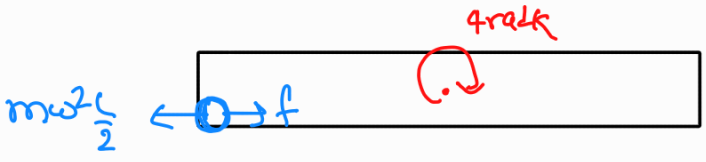
$$= 8 \times \frac{\pi}{4} = 2\pi =$$

(37) time taken by insects to reach the other end,

$$t_0 = \frac{\frac{1}{2}m}{\frac{1}{2}m\omega} = 1\text{Sec}$$

angular velocity at that instant

$$\omega = \frac{8}{1+t^2} = \frac{8}{1+1} = 4\ \text{rad/s}$$



$$f = m\omega^2\frac{L}{2} = \frac{1}{6} \times 4^2 \times \frac{1}{2} = \frac{4}{3}N$$

(38) EMF at time t

$$\mathcal{E}(t) = \frac{1}{2}B\omega a^2$$

current, $i(t) = \frac{\mathcal{E}(t)}{R} = \frac{B\omega a^2}{2R}$

From work Energy theorem

$$W_R = \Delta K + \Delta P$$

$$-i^2Rt = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 - mgx$$

Diff wrt time

$$-i^2R = mv\frac{dv}{dt} + I\omega\frac{d\omega}{dt} - mg\frac{dx}{dt}$$

we know, $v = \omega a$

$$\frac{dv}{dt} = a\frac{d\omega}{dt}$$

$$-\left(\frac{B\omega a^2}{2R}\right)^2 R = mv\frac{dv}{dt} + \frac{m}{2} \times \frac{v}{a} \times \frac{1}{a} \times \frac{dv}{dt} - mgv$$

$$-\frac{B^2 v^2 a^2}{4R} = mv\frac{dv}{dt} + \frac{mv}{2}\frac{dv}{dt} - mgv$$

$$\frac{3mv}{2}\frac{dv}{dt} - mgv + \frac{B^2 v^2 a^2}{4R} = 0$$

$$\frac{3m}{2}\frac{dv}{dt} + \frac{B^2 a^2 v}{4R} - mg = 0$$

$$\frac{3m}{2}\frac{dv}{dt} = mg - \frac{B^2 a^2}{4R} v$$

$$\frac{dv}{dt} = \frac{2g}{3} - \frac{B^2 a^2}{6mR} v$$

$$\int_0^v \frac{dv}{\frac{2g}{3} - \frac{B^2 a^2}{6mR} v} = \int_0^t dt$$

$$\frac{\ln\left[\frac{2g}{3} - \frac{B^2 a^2}{6mR} v\right]_0^v}{-\frac{B^2 a^2}{6mR}} = [t]_0^t$$

$$\ln\left[\frac{\frac{2g}{3} - \frac{B^2 a^2}{6mR} v}{2g/3}\right] = -\frac{B^2 a^2 t}{6mR}$$

$$1 - \frac{B^2 a^2}{4mgR} v = e^{-\frac{B^2 a^2 t}{6mR}}$$

$$\frac{B^2a^2}{4mgR}v = 1 - e^{-\frac{B^2a^2t}{6mR}}$$

$$v = \frac{4mgR}{B^2a^2}\left(1 - e^{-\frac{B^2a^2t}{6mR}}\right)$$

for terminal velocity, $t \to \infty$

$$v_{terminal} = \frac{4mgR}{B^2a^2}$$

$$a = \frac{dv}{dt} = \frac{d}{dt}\left[\frac{4mgR}{B^2a^2}\left(1 - e^{-\frac{B^2a^2t}{6mR}}\right)\right]$$

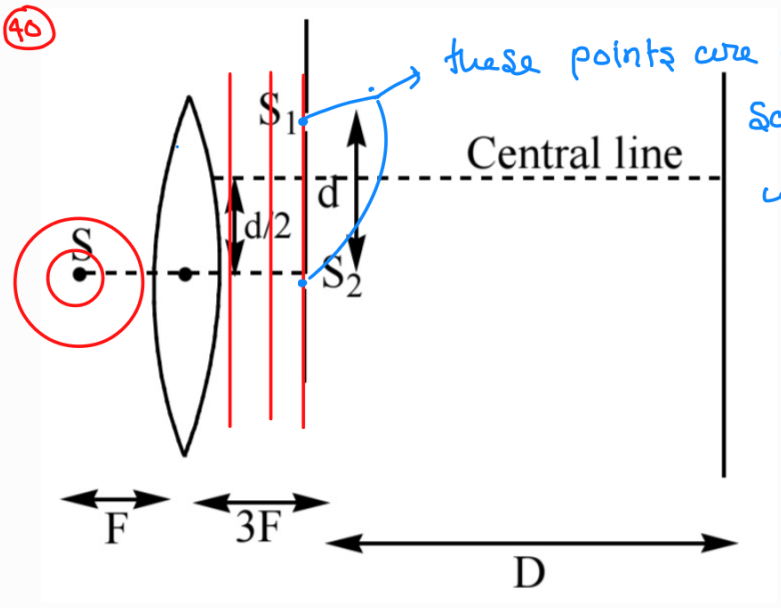
$$= \frac{4mgR}{B^2a^2}\left(-e^{-\frac{B^2a^2t}{6mR}} \times -\frac{B^2a^2}{6mR}\right)$$

$$= \frac{2g}{3}e^{-\frac{B^2a^2t}{6mR}}$$

at $t = 0$

$$a = \frac{2g}{3}$$

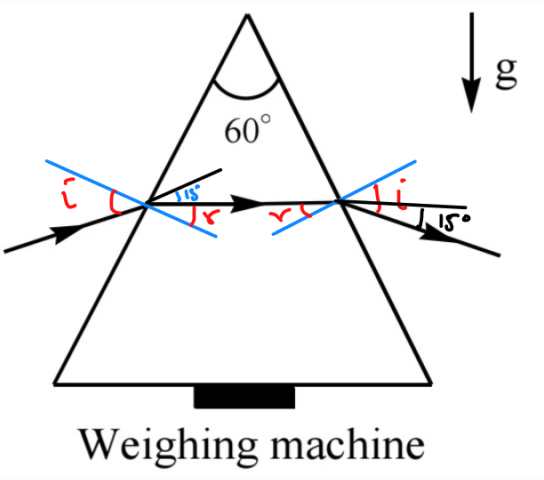
(40)



these points are on same wave front so they are in same phase. thus central bright fringe will not shift.

$\Rightarrow \mu_1 = 0$

(41)



as ray is parallel to base it is case of minimum deviation

$\Rightarrow r_1 = r_2 = r = \frac{A}{2} = \frac{60}{2} = 30°$

$1 \times \sin i = \mu \sin r$

$\sin i = \sqrt{2} \sin 30 = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$

$i = 45°$

$F = \frac{P}{C} \sin 15°$

on 2nd refraction

$F_{total} = \frac{2P}{C} \sin 15°$

$= \frac{2P_0}{C} \left(\frac{\sqrt{3}-1}{\sqrt{2}} \right)$

(42) For A & block

$(60+5) \times 2 = 60v_1 + 5v$

$60v_1 = 130 - 5v$

$v_1 = \frac{130 - 5v}{60}$ - (1)

For B & Block

$(60 \times 1) + 5V = (60+5)V_2$

$$V_2 = \frac{60+5V}{65} \quad -(11)$$

For avoiding collision

$V_1 = V_2$

$$\frac{130-5V}{\cancel{60}\ 12} = \frac{60+5V}{\cancel{65}\ 13}$$

$1690 - 65V = 720 + 60V$

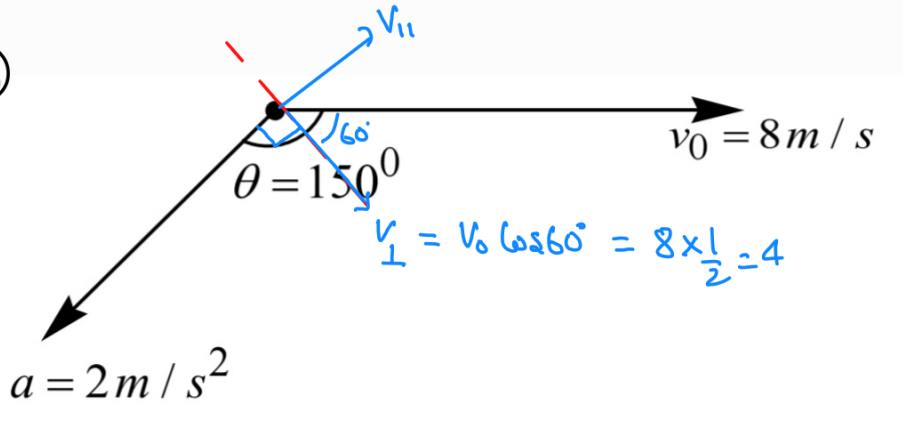
$125V = 970$

$$V = \frac{970}{125}$$

$$n + \frac{76}{100} = \frac{970}{125}$$

$$n = \frac{970}{125} - \frac{76}{100} = \frac{3880 - 380}{500} = \frac{3500}{500} = 7$$

(93)



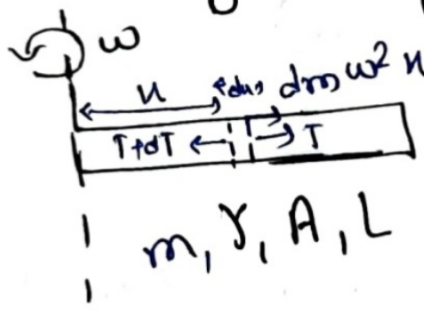
$V_\perp = V_0 \cos 60^\circ = 8 \times \frac{1}{2} = 4$

$a_c = \frac{V_\perp^2}{R}$

$R = \frac{(V_\perp)^2}{a}$

$= \frac{4^2}{2} = 8\ m$

(44) Find Elongation of rod.



tension in elementary,

$dT = dm\,\omega^2 x$

$= \left(\frac{m}{L}dx\right)\omega^2 x$

$\int dT = \frac{m\omega^2}{L}\int_x^L x\,dL$

$T = \frac{m\omega^2}{2L}(L^2 - x^2)$

Elongation in elementary

$d(\Delta L) = \frac{T\,dx}{AY}$

$= \frac{m\omega^2(L^2-x^2)}{2L \times AY}dx$

$\Delta L_1 = \frac{m\omega^2}{2ALY}\int_0^{L/2}(L^2-x^2)dx$

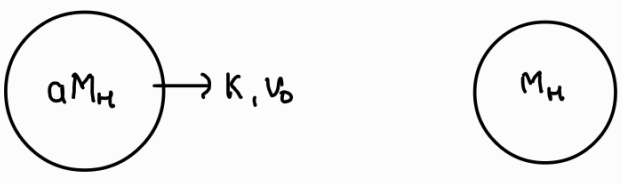
$\Delta L_2 = \frac{m\omega^2}{2ALY}\int_{L/2}^{L}(L^2-x^2)dx$

$\frac{\Delta L_1}{\Delta L_2} = \frac{\left[L^2x - \frac{x^3}{3}\right]_0^{L/2}}{\left[L^2x - \frac{x^3}{3}\right]_{L/2}^{L}} = \frac{\frac{L^3}{2} - \frac{L^3}{24}}{L^3 - \frac{L^3}{3} - \frac{L^3}{2} + \frac{L^3}{24}} = \frac{\left(\frac{12-1}{24}\right)}{\frac{24-8-12+1}{24}}$

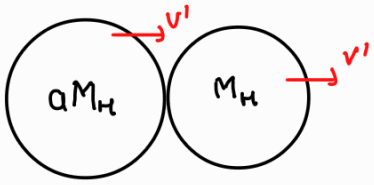
$= \frac{11}{5}$

$\frac{\Delta L_2}{\Delta L_1} = \frac{5}{11}$

(45)



For minimum K



$$aM_H V_0 = (aM_H + M_H) V'$$

$$V' = \frac{aV_0}{a+1}$$

$$K_f = \frac{1}{2}(a+1) M_H V'^2 = \frac{1}{2}(a+1) M_H \times \frac{a^2 V_0^2}{(a+1)^2}$$

$$= \left(\frac{1}{2} aM_H V_0^2\right) \times \frac{a}{a+1} = \frac{a}{a+1} K$$

$$\Delta K = K - \frac{aK}{a+1} = \frac{K}{a+1}$$

$$13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{51}{a+1}$$

$$13.6 \times \frac{3}{4} = \frac{51}{a+1}$$

$$a+1 = \frac{51 \times 4}{13.6 \times 3} = 5$$

$$a = 5 - 1 = 4$$

$\Rightarrow$ $Z = 2$ $\rightarrow$ It is Helium.

Energy released from He^{2+} in transition of $n=4$ to $n=2$

$$E = 13.6\left(\frac{1}{2^2} - \frac{1}{4^2}\right) \times 2^2 \text{ eV}$$

$$= 13.6 \times \left(\frac{4-1}{16}\right) \times 4 \text{ eV}$$

$$= 13.6 \times \frac{3}{\cancel{16}\,4} \times \cancel{4} \text{ eV}$$

$$= 3.4 \times 3 \text{ eV}$$

$$= 10.2 \text{ eV}$$

on Photoelectric effect

$KE_{max} = 4eV$

$KE_{max} = 10.2 - \phi$

$4 = 10.2 - \phi$

$\phi = 10.2 - 4 = 6.2$

$\frac{1240}{\lambda} = 6.2$

$\lambda = \frac{1240}{6.2}\,nm = 200\,nm$

$x = 2$

(46) $Q = \left[\left(M_N + M_{He}\right) - \left(M_H + M_O\right)\right] C^2$

$= \left[(16.006 + 4.003) - (1.008 + 19.003)\right] \times 930\ MeV$

$= (20.009 - 20.011) \times 930\ MeV$

$= -0.002 \times 930\ MeV$

$= -1.860\ MeV$

$= -1.86\ MeV$

$K_{min} = \frac{m_N + m_{He}}{m_{He}} Q$

$= \frac{16 + 4}{4} \times 1.86\ MeV$

$= 5 \times 1.86\ MeV$

$= 9.30\ MeV \approx 9\ MeV.$